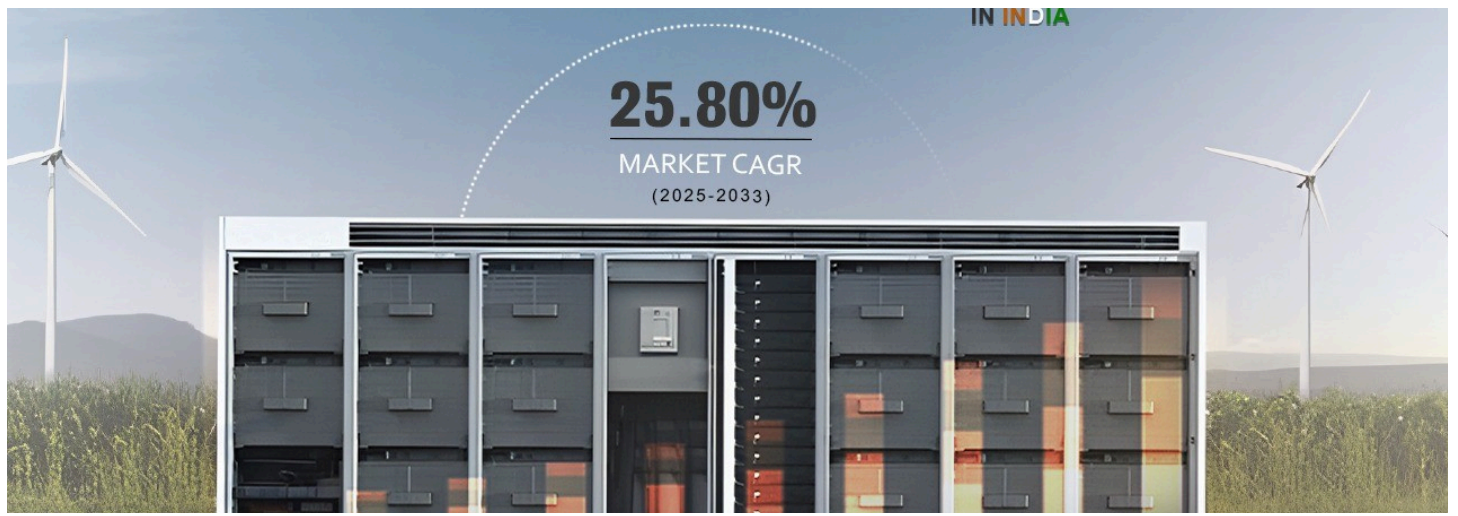




Demand Forecast for Battery Energy Storage Systems in India



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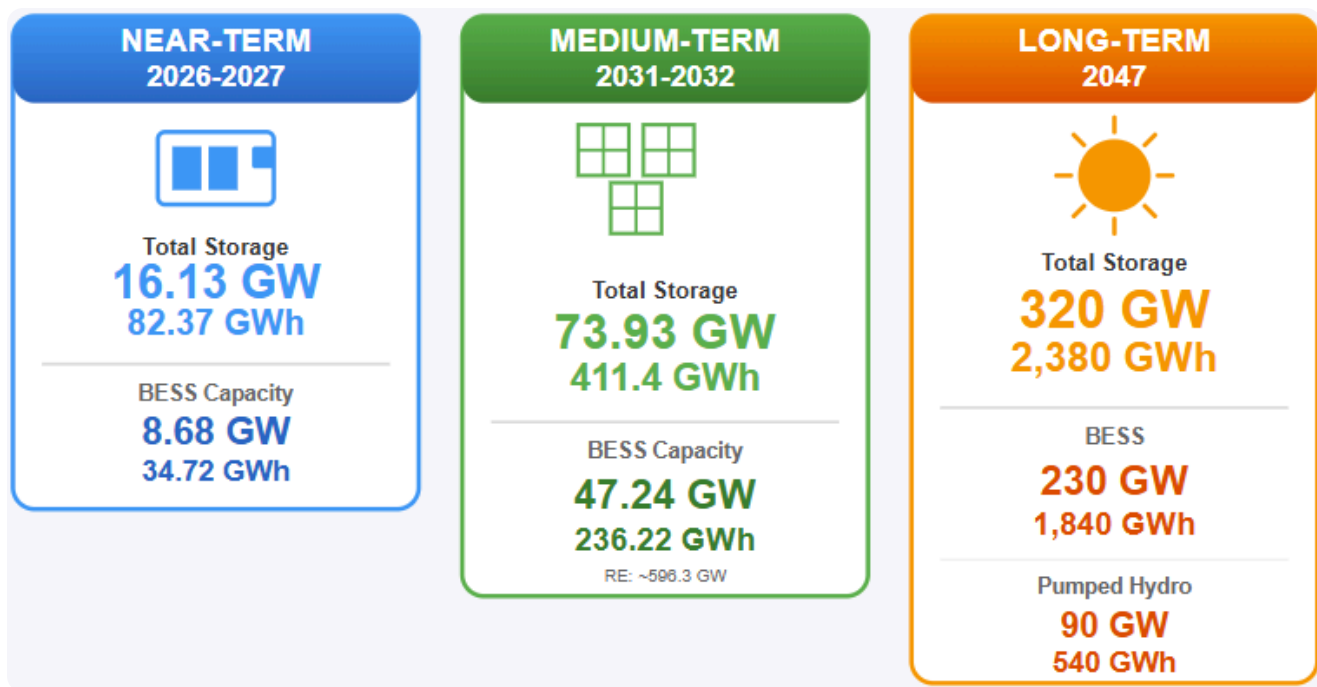
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Demand Forecast for Battery Energy Storage Systems in India

The Battery Energy Storage Systems (BESS) market in India stands at a critical inflection point, with government targets and market dynamics converging to create unprecedented growth opportunities. As India progresses toward its ambitious renewable energy goals, the demand for BESS has transformed from a secondary consideration into a core component of the nation's energy infrastructure strategy.

Government Targets and Capacity Requirements

The **Central Electricity Authority (CEA)** has established the most definitive capacity targets for energy storage deployment in India. According to the National Electricity Plan 2023, India requires progressively increasing storage capacity to integrate its renewable energy sources effectively:

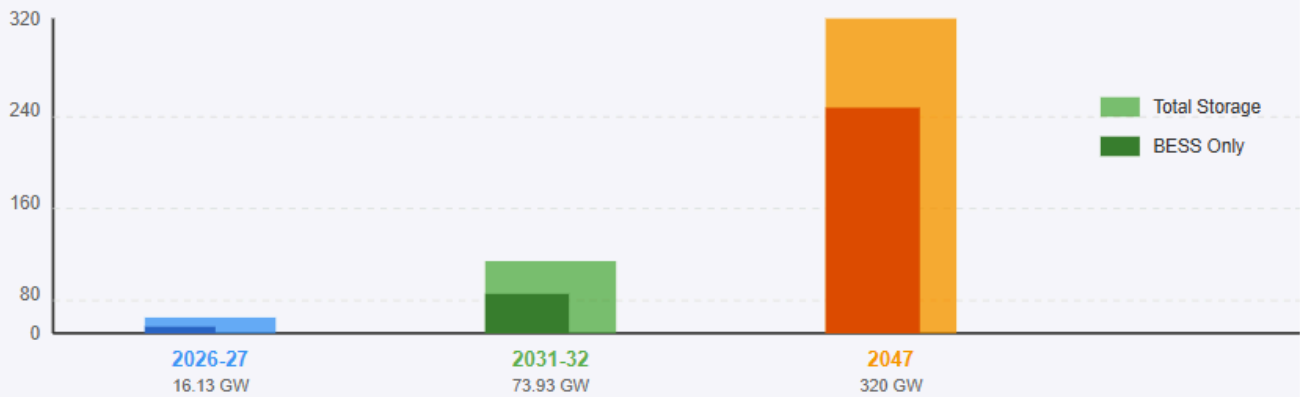


- **Near-term Targets (2026-2027):** India needs approximately **16.13 GW / 82.37 GWh** of total energy storage capacity, of which **8.68 GW / 34.72 GWh** is specifically BESS. This represents the baseline requirement for managing renewable energy intermittency as India scales its solar and wind installations.
- **Medium-term Targets (2031-2032):** The capacity requirement escalates dramatically to **73.93 GW / 411.4 GWh** of total energy storage, with **47.24 GW / 236.22 GWh** being BESS. This substantial increase reflects the anticipated growth of renewable installations to approximately 596.3 GW by this period.
- **Long-term Vision (2047):** Projections extend further, indicating India could require **320 GW / 2,380 GWh** of total storage capacity, comprising **230 GW / 1,840 GWh** from BESS and **90 GW / 540 GWh** from pumped hydro storage.

Key Insights

- **Near-term (2026-27):** Baseline capacity to manage renewable intermittency
- **Medium-term (2031-32):** 4.6x increase supporting ~596.3 GW renewable capacity
- **Long-term (2047):** 20x growth from baseline, combining BESS and pumped hydro
- BESS dominates storage mix, providing flexibility and faster response times

Capacity Growth Comparison (GW)

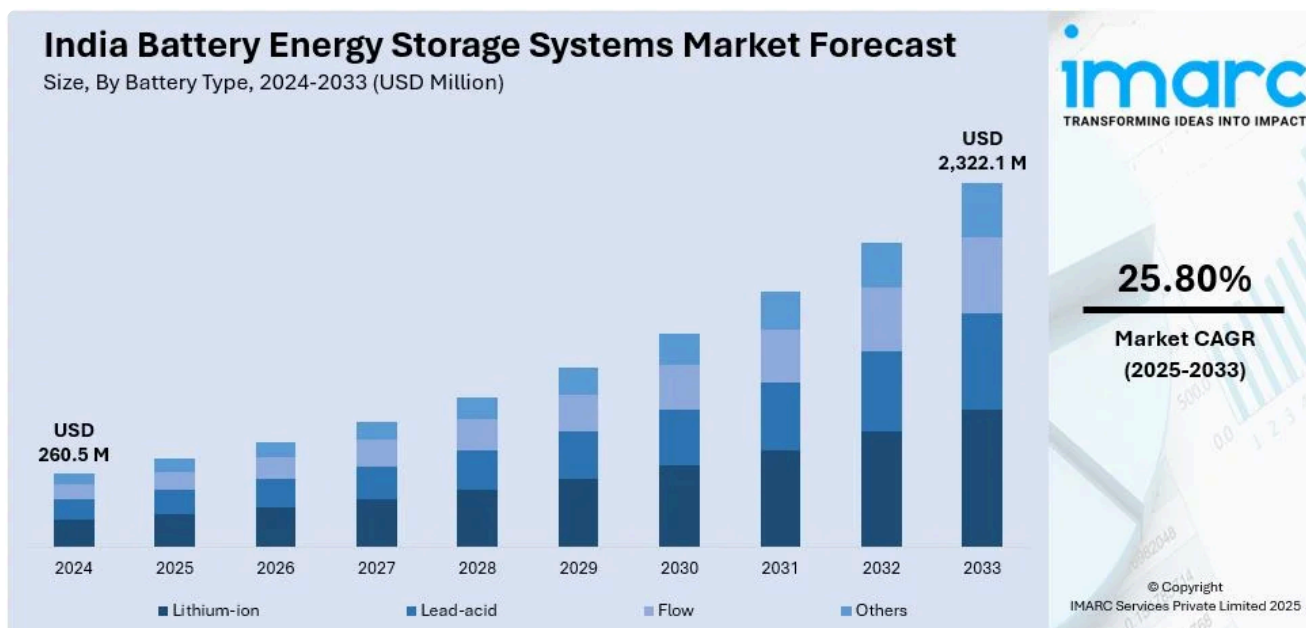


Different research organizations provide varying market forecasts, though all point toward significant growth. Some projections suggest India will need as little as **61 GW** by 2030 for core grid stability, while more aggressive assessments estimate requirements approaching **74 GW** by 2032 to fully integrate the renewable energy capacity.

Market Size and Growth Trajectory

The financial scale of the BESS market reflects its strategic importance. India's BESS market was valued at approximately **USD 260.5 million to USD 3.5 billion in 2023-2024**, depending on the scope and definition employed by different research organizations. By 2030, market projections range widely:

- IMARC Group projects the market reaching **USD 2,322.1 million by 2033**, with a compound annual growth rate (CAGR) of **25.80%** during 2025-2033.



- Grand View Research forecasts the market will reach **USD 5,318.2 million by 2030**, with an aggressive CAGR of **40.9%** from 2024-2030.
- Alternative estimates suggest valuations between USD 9 billion to USD 32 billion by 2030-2033, reflecting the market's potential range.

This growth translates to capital investments exceeding **₹3.5 lakh crore (approximately USD 42 billion)** between 2022 and 2032 for BESS deployment alone.

Current Deployment Status and Pipeline

As of October 2025, India's operational BESS capacity stands at only **0.5 GWh**, highlighting the market's nascent stage despite significant policy momentum. However, the tender pipeline reveals substantial growth anticipated in the coming years.

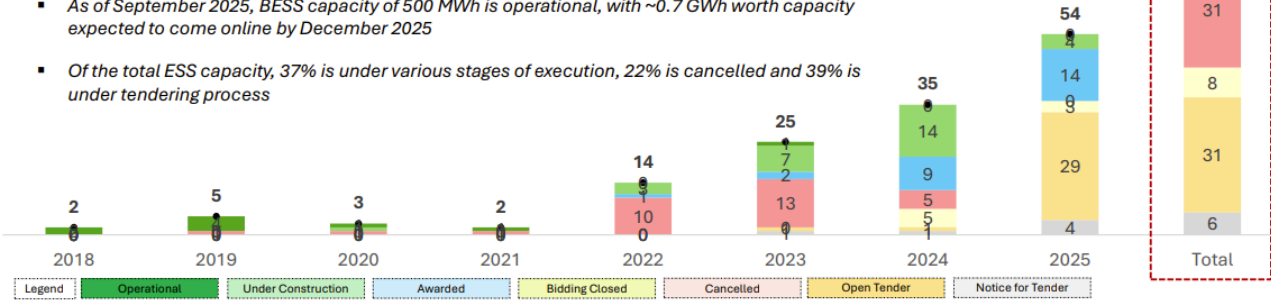
Current Tender Status (as of October 2025):

- **Total ESS capacity tendered:** 195 GWh (comprising both BESS and pumped storage hydro)
- **BESS capacity specifically:** 80 GWh
- **Projects under execution:** 17 GWh of BESS
- **Projects in tendering process:** 25.41 GWh of BESS
- **Cancelled tenders:** 8.22 GWh of BESS



Stage	Number of Tenders	RE Capacity	ESS Capacity	PSP Capacity	BESS Capacity
		GW	GWh	GWh	GWh
NIT	6	1.21	2.21	2.00	0.21
RfS	31	5.78	51.66	26.25	25.41
Bidding Closed	8	1.45	18.08	12.00	6.08
Cancelled	31	11.43	42.72	34.63	8.22
Awarded	26	12.32	23.28		23.28
Under Construction	29	18.16	56.04	39.45	16.59
Operational	9	0.86	0.50	0.00	0.50
Total	140	51	195	114	80

- As of September 2025, BESS capacity of 500 MWh is operational, with ~0.7 GWh worth capacity expected to come online by December 2025
- Of the total ESS capacity, 37% is under various stages of execution, 22% is cancelled and 39% is under tendering process



Note: The numbers in the graph refers to the number of tenders. Captures only projects/ tenders in competitive bidding domain
Source: Tender nodal agency websites, press releases, compiled by Debmalya Sen

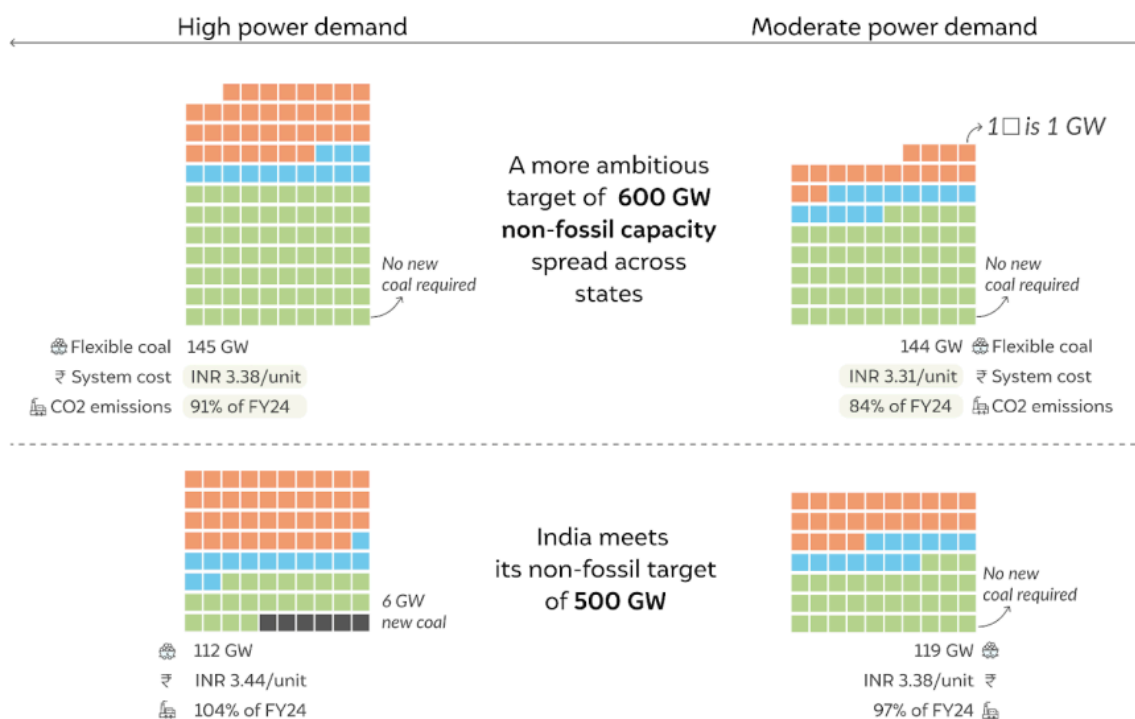
Updated till 30 October 2025

The significant shift from earlier years is evident: **zero cancellations occurred in 2025** compared to substantial cancellations in prior years, indicating improved policy certainty and market viability.

Key Growth Drivers

Renewable Energy Integration remains the primary driver of BESS demand. With India targeting **500 GW of non-fossil fuel capacity by 2030** and potentially **600 GW** to meet accelerating electricity demand, storage solutions are essential for managing the intermittency of solar and wind power.

A 600-GW RE pathway will help India meet rising power demand most reliably and affordably



Additional resources required

New coal
 Battery energy storage system
 Pumped storage hydro
 Interregional transmission
 Interstate transmission

Note: Power sector CO2 emissions in FY24 = 1,260 MT (Niti Aayog, n.d.)

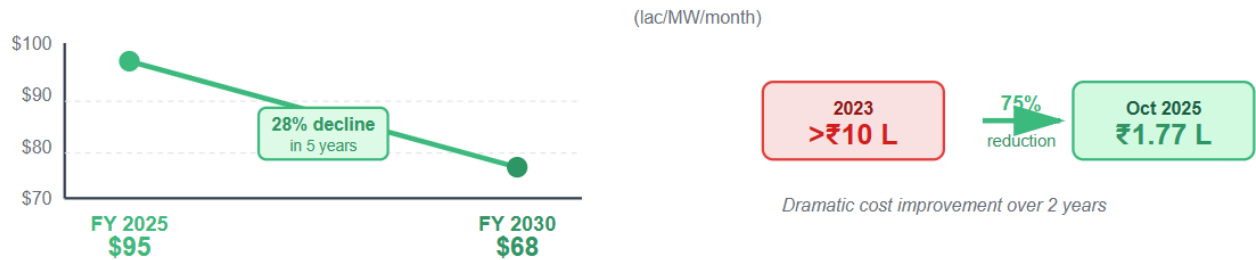
Source: How can India Meet its Rising Power Demand? (2025)



- Declining Battery Costs** substantially improve BESS economics. Global lithium-ion battery pack prices are expected to decline from approximately **USD 95/kWh in FY 2025 to USD 68/kWh in FY 2030**. Domestically, tariff discovery for standalone BESS has dropped **75% over two years**, falling from rates exceeding 10 lac/MW/month to as low as 1.77 lac/MW/month by October 2025.

1. Declining Battery Costs

Global Lithium-Ion Battery Pack Prices (USD/kWh) Domestic BESS Tariff Discovery



- **Government Support Mechanisms** accelerate deployment through the Viability Gap Funding (VGF) scheme. The first phase of VGF (providing 30% subsidy) already supports **13.2 GWh** across 17 tenders, while VGF 2 (providing 16% support) targets an additional **30 GWh**. The government aims to develop **4,000 MWh of BESS projects by 2030-31** through these schemes alone.

2. Government Support Mechanisms



- **Electric Vehicle Ecosystem Growth** creates additional BESS demand beyond grid-scale applications. The expansion of EV charging infrastructure requires battery storage to manage peak loads, creating complementary market opportunities.
- **Hybrid Renewable Projects** increasingly incorporate mandatory BESS components. Government mandates requiring 5% of renewable projects to include storage with at least two-hour capacity are driving hybrid solar-wind-BESS integration at scale.

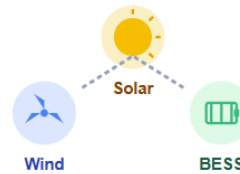
3. EV Ecosystem Growth



Charging Infrastructure

- Peak load management
- Grid stability support
- Complementary market opportunities

4. Hybrid Renewable Projects



Government Mandate

5% of renewable projects must include storage with ≥2 hour capacity

Converging Forces Accelerating India's BESS Revolution

Economic viability + Policy support + Market expansion + Technology integration

Creating a sustainable foundation for large-scale energy storage deployment

75%

Tariff Reduction
(2023-2025)

43.2 GWh

VGf Support
(Phase 1 + 2)

28%

Battery Cost Drop
(2025-2030)

5%

Hybrid Mandate
(≥2 hr storage)

Technology and Market Segmentation

Utility-Scale Applications dominate the market, with large-scale renewable projects incorporating BESS for grid stability. As of 2023, utility-scale storage represented the largest revenue-generating segment, a position expected to strengthen as grid-integration requirements intensify.

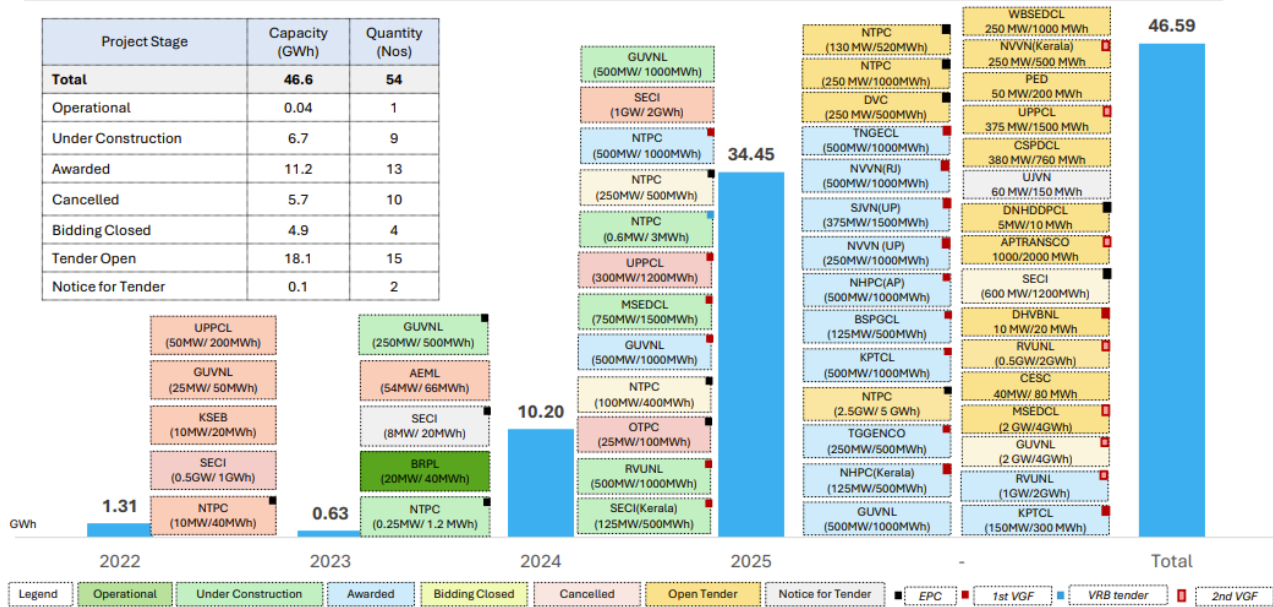
Tender Categories and Applications:

- **Standalone BESS: 34.6 GWh tendered (2025)**

Standalone BESS Tenders Growth Trend | 2022 - 2025



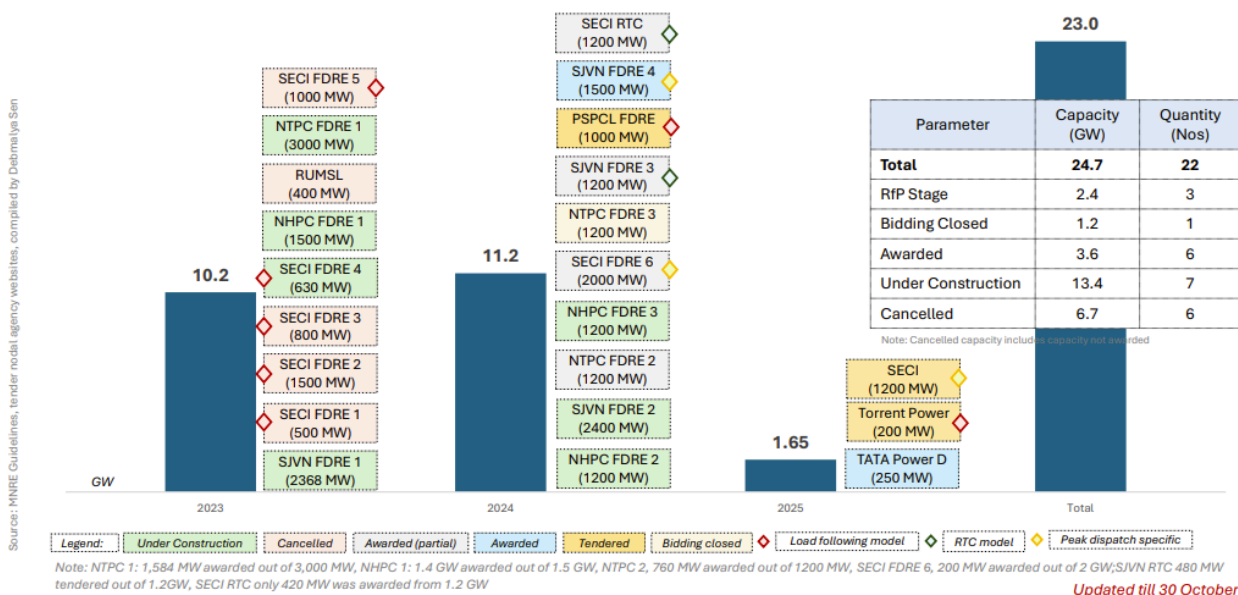
VGf has transformed the standalone BESS market, VGf 1 (30% Support) for 13.2 GWh (17 tenders) has already been completed. VGf 2 (16% support) to support 30 GWh is active now with already 16 GWh (7 number) worth tenders out



- **Firm Dispatchable Renewable Energy (FDRE) with storage: 23 GWh tendered**

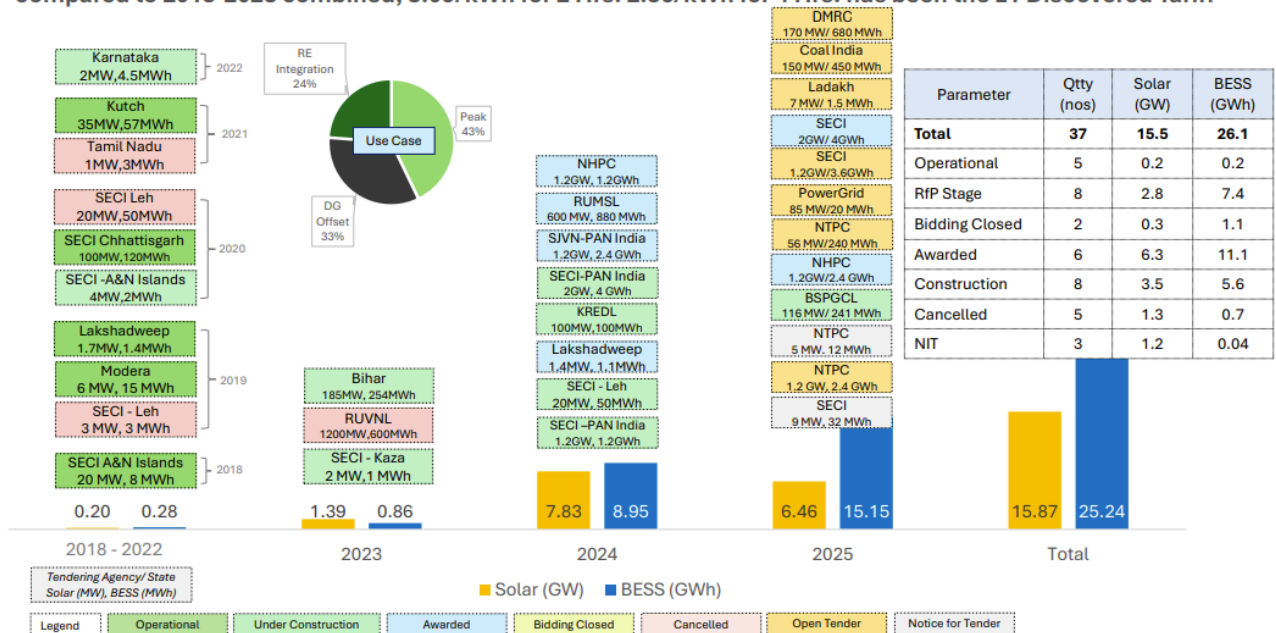
Firm and Dispatchable Renewables (FDRE) Status Overview | 2023 - 2025

FDRE consists of 4 different configurations, including (1) **assured peak** power, (2) Round the Clock (RTC) RE with firm delivery during demand hours, (3) **Load following** power delivery as specified by DISCOM, and (4) RE projects with **Firm delivery of power for fixed hours** as per requirement of DISCOMs.



- Solar+BESS hybrid projects: 15.2 GWh tendered

Solar with BESS gained significant traction in 2024, the year saw 6.5X increase in tendered capacity compared to 2018-2023 combined, 3.09/kWh for 2 Hrs. 2.86/kWh for 4 Hrs. has been the L1 Discovered Tariff



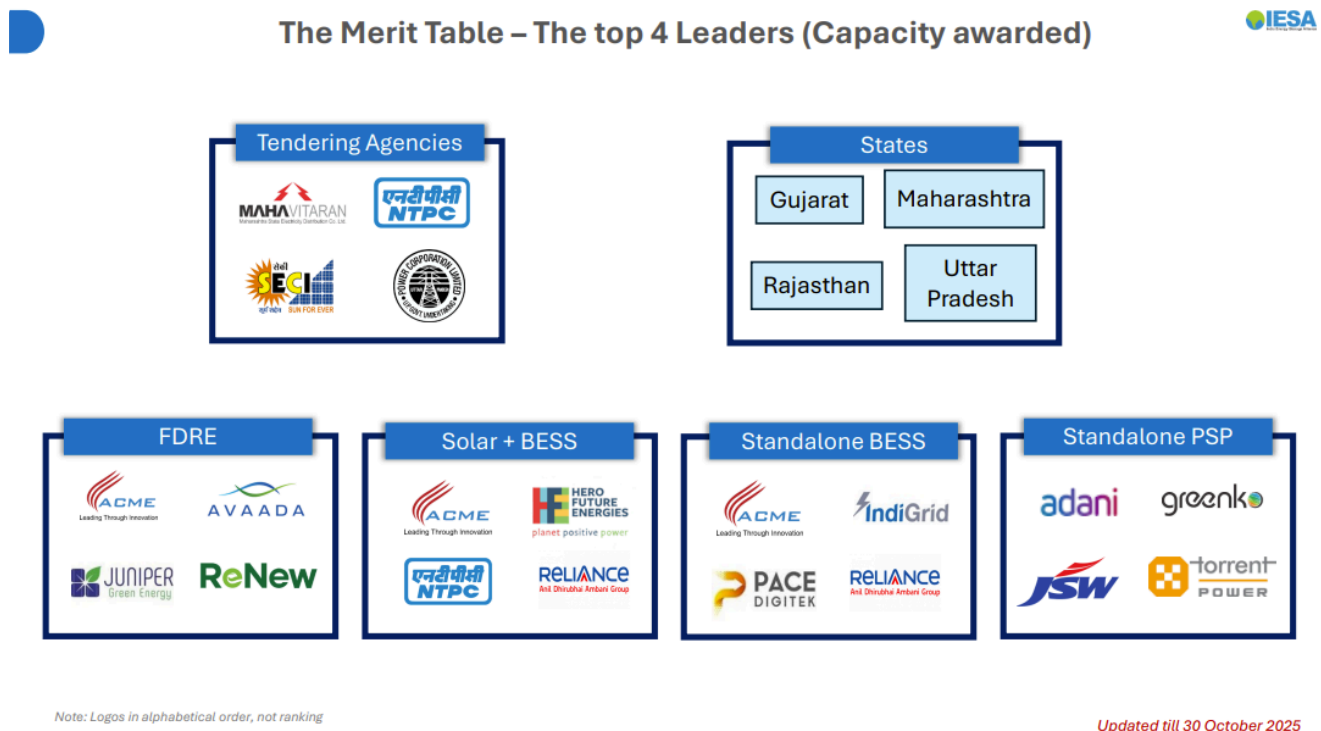
Within FDRE projects, multiple dispatch models are emerging: round-the-clock (RTC) delivery, peak assurance, load-following capability, and peak-specific dispatch, each addressing distinct grid stability requirements.

Commercial and Industrial Segments are emerging as secondary growth areas, driven by the need for demand-side management and energy arbitrage opportunities. Energy arbitrage potential is estimated at ₹2.5-₹3 per kWh by 2025, creating revenue streams beyond capacity payments.

Geographic Distribution and State-Level Growth

Leading states in BESS deployment include Gujarat, Rajasthan, Uttar Pradesh, and Maharashtra, which together account for the majority of awarded and operational capacity. State-level tenders reveal a geographic shift toward more distributed deployment as states increasingly recognize BESS's role in local grid stability.

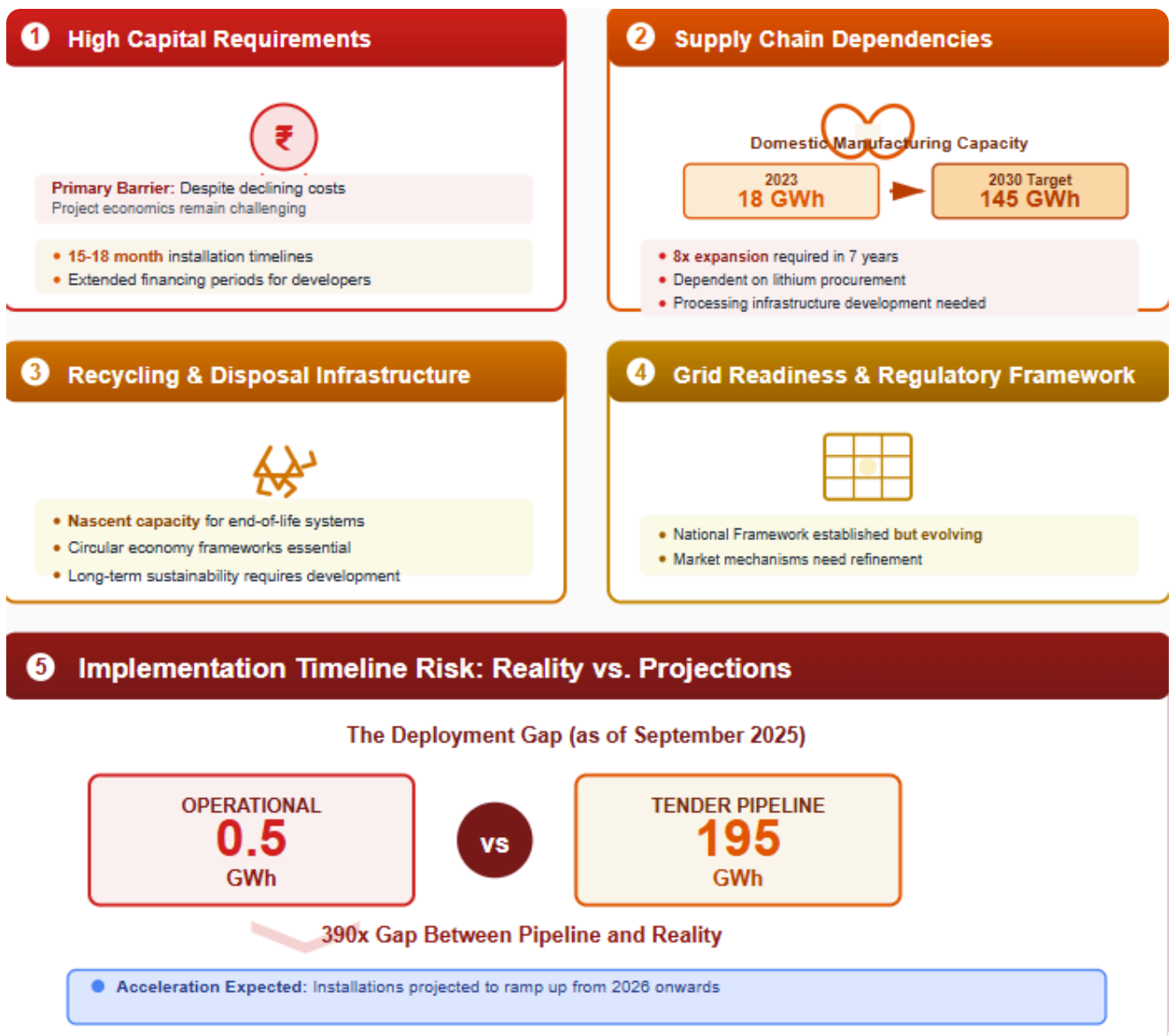
As of October 2025, active tenders are distributed across virtually all major states, with significant pipeline capacity in West Bengal (1,000 MWh), Tamil Nadu (1,500 MWh), Haryana, Andhra Pradesh, and Bihar, indicating a move beyond traditional renewable energy hubs.



Challenges to Achieving Demand Forecasts

Despite optimistic projections, several obstacles could slow BESS deployment:

- **High Capital Requirements** remain the primary barrier, though declining battery costs are progressively reducing project economics. Installation timelines extending 15-18 months create financing challenges for project developers.
- **Supply Chain Dependencies** on lithium and other critical materials create vulnerability. While India's domestic lithium-ion battery manufacturing capacity was 18 GWh in 2023, expansion to 145 GWh by 2030 remains ambitious and dependent on successful procurement and processing infrastructure development.
- **Recycling and Disposal Infrastructure** remains nascent, with limited capacity to handle end-of-life battery systems. Developing circular economy frameworks is essential for long-term sustainability.



- **Grid Readiness and Regulatory Framework** for energy storage integration continues evolving. While the National Framework for Promoting Energy Storage Systems was established, implementation of market mechanisms for storage participation and ancillary services pricing requires further refinement.
- **Implementation Timeline Risk:** Current ground installations lag projections. As of September 2025, only 0.5 GWh is operational despite 195 GWh in the tender pipeline. However, analyst estimates suggest installations will accelerate from 2026 onwards, potentially reaching **~50 GWh by 2030** if projects currently in execution complete on schedule.

Market Outlook and Investment Implications

India's BESS market exhibits characteristics of rapid transition from demonstration phase to large-scale deployment. The convergence of declining costs, government support, and binding renewable energy targets creates strong structural demand growth. Between 2025 and 2030, India is positioned to deploy approximately **50 GWh of BESS**, approaching or potentially exceeding near-term CEA targets if execution challenges are overcome.

The critical inflection point occurs during 2026-2027, when multiple large, tendered projects are scheduled for commissioning and when tariff discovery trends stabilize. Successfully executing the current pipeline of 17 GWh under construction while progressing 25 GWh in active tenders will determine whether India achieves its goal of 236-411 GWh BESS capacity by 2032.

For stakeholders in India's energy sector, the BESS demand forecast represents not just a storage opportunity but a fundamental requirement for achieving renewable energy ambitions and maintaining grid reliability as India's electricity system undergoes its most significant transformation in decades.

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